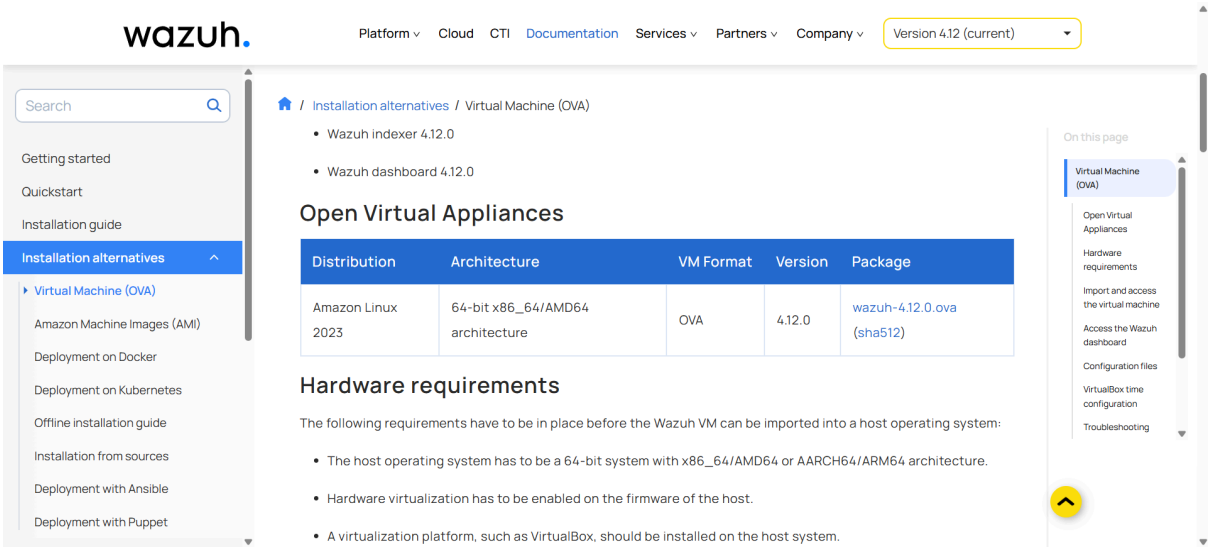
Projet Labs IDS (Wazuh et Suricata)

J'ai télécharger la VM wazuh-4.14.4 depuis le site de Wazuh :

https://documentation.wazuh.com/current/deployment-options/virtual-machine/virtual-machine.html



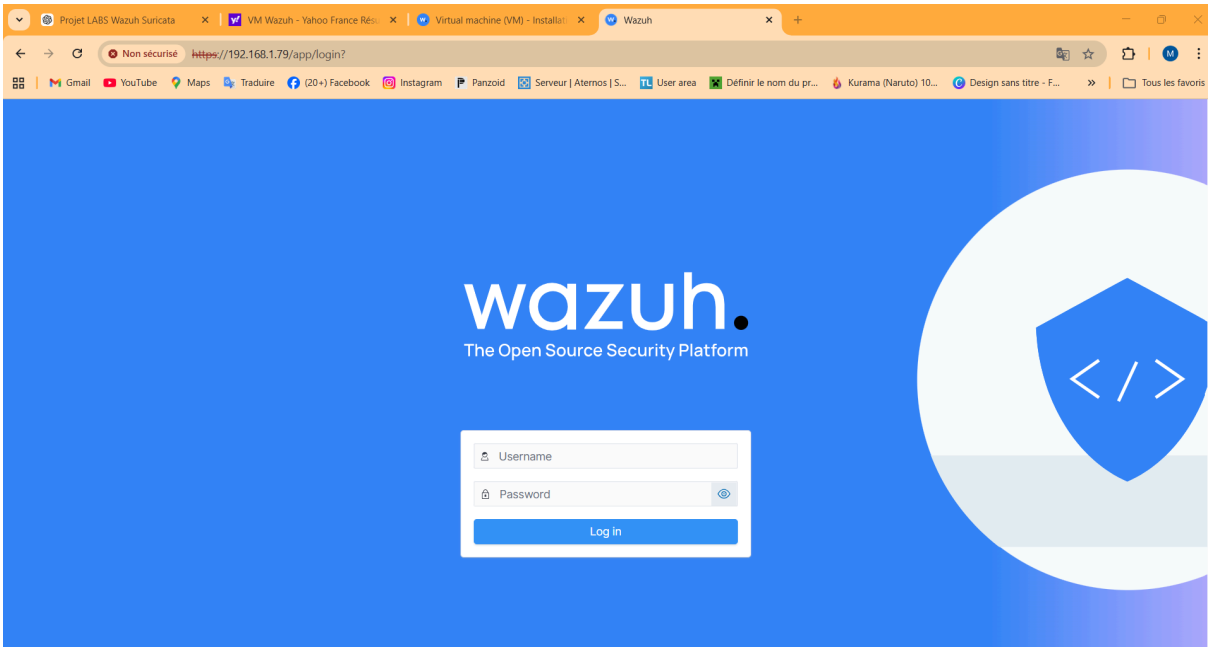
user: wazuh-user
password: wazuh

SSH root user login has been deactivated; nevertheless, the wazuh-user retains sudo privileges. Root privilege escalation can be achieved by executing the following command:

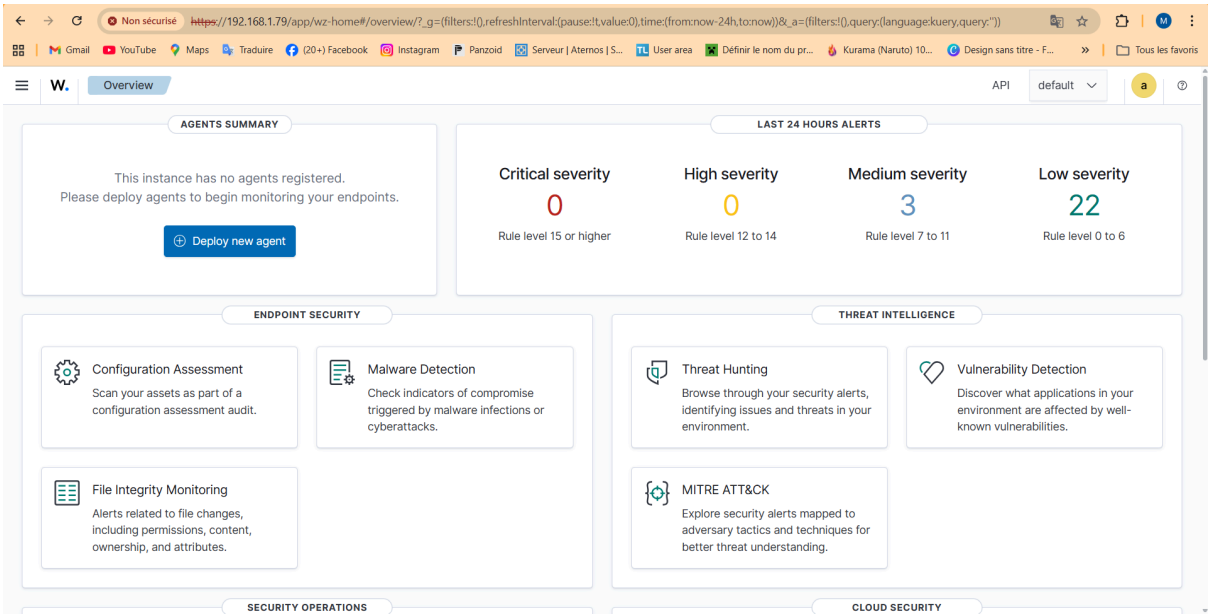
```
sudo -i
```

URL: https://<wazuh_server_ip>
user: admin
password: admin

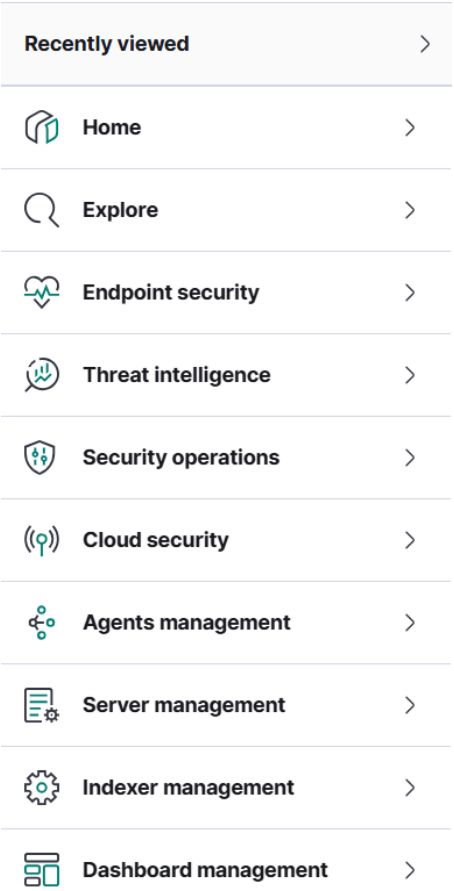
Page de connexion Wazuh :



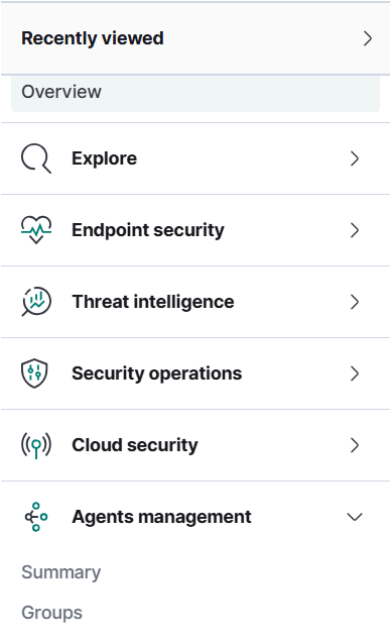
Interface de Wazuh :



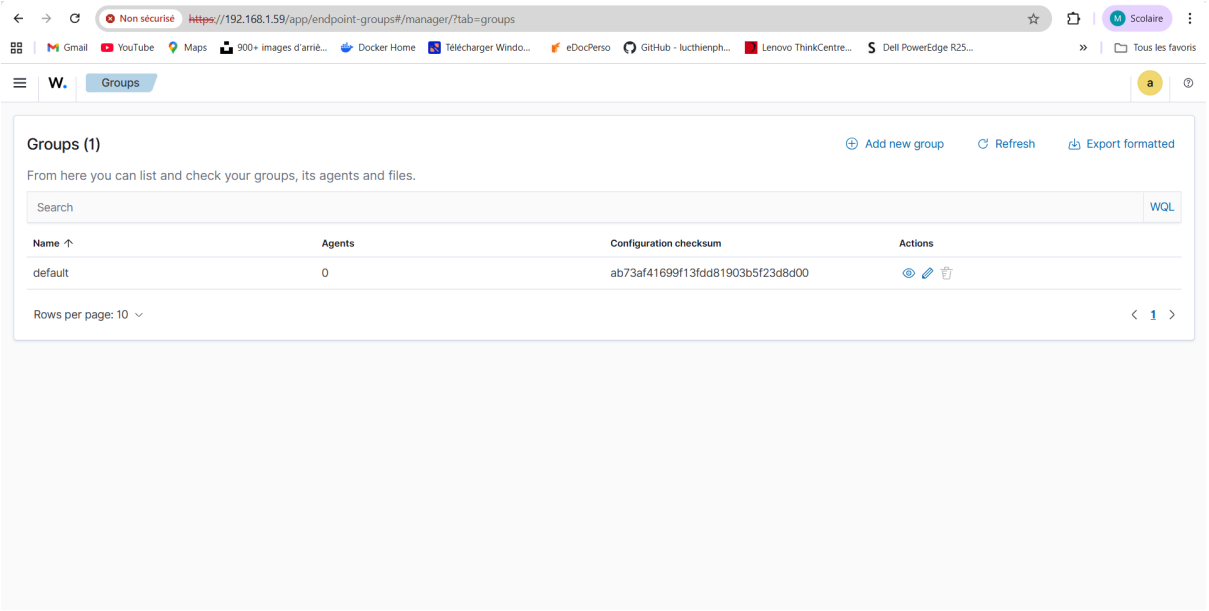
Les options de Wazuh :



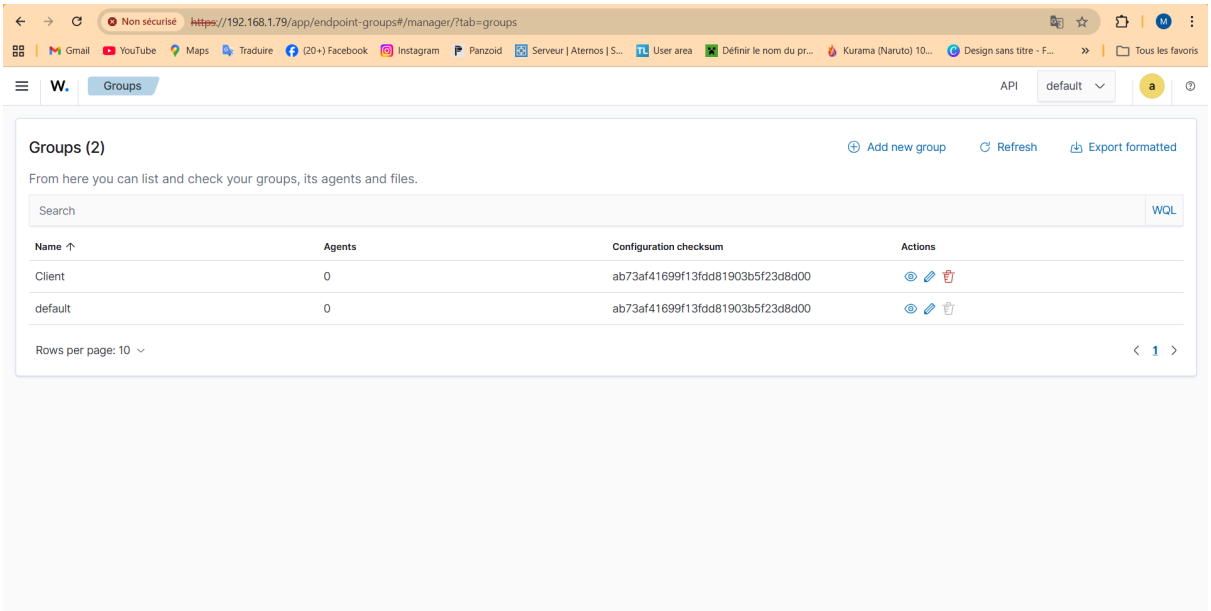
Créer des groupes sur Wazuh :



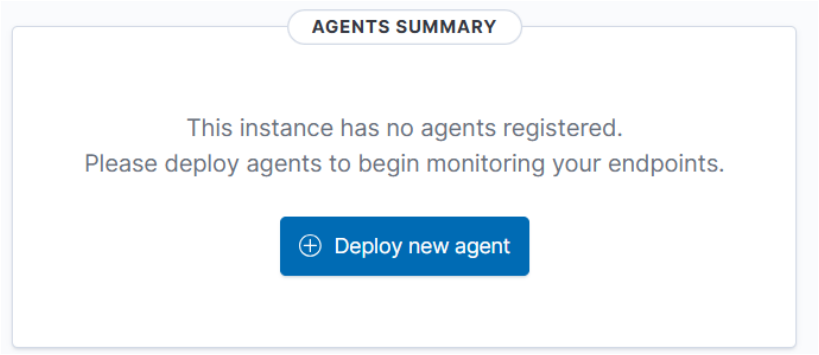
Interface pour créer des groupes :

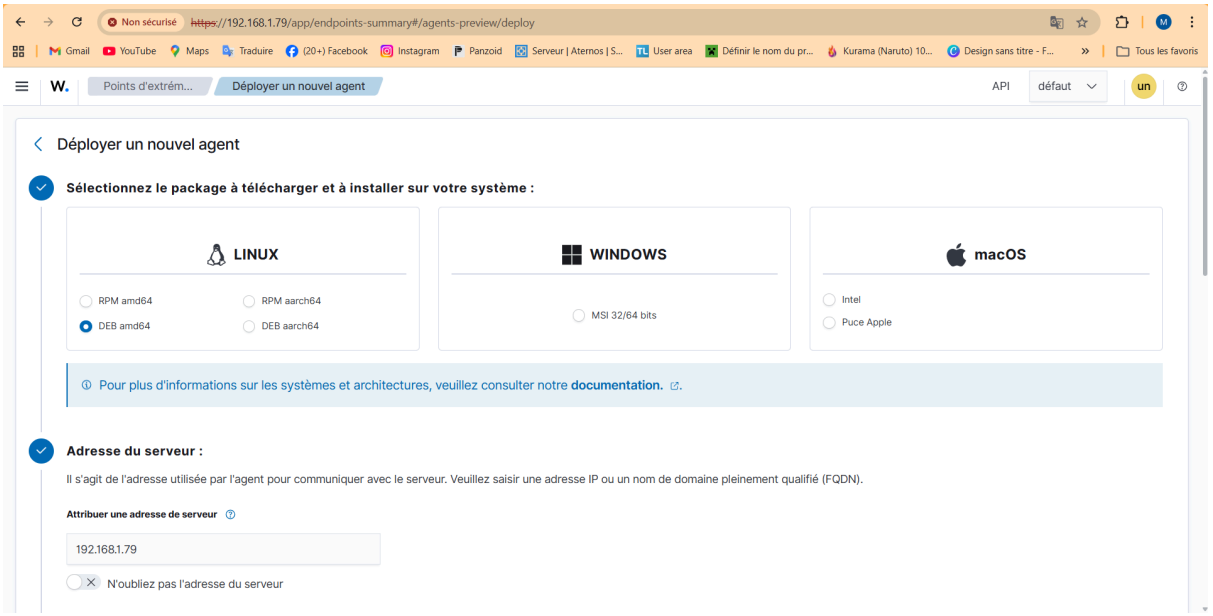


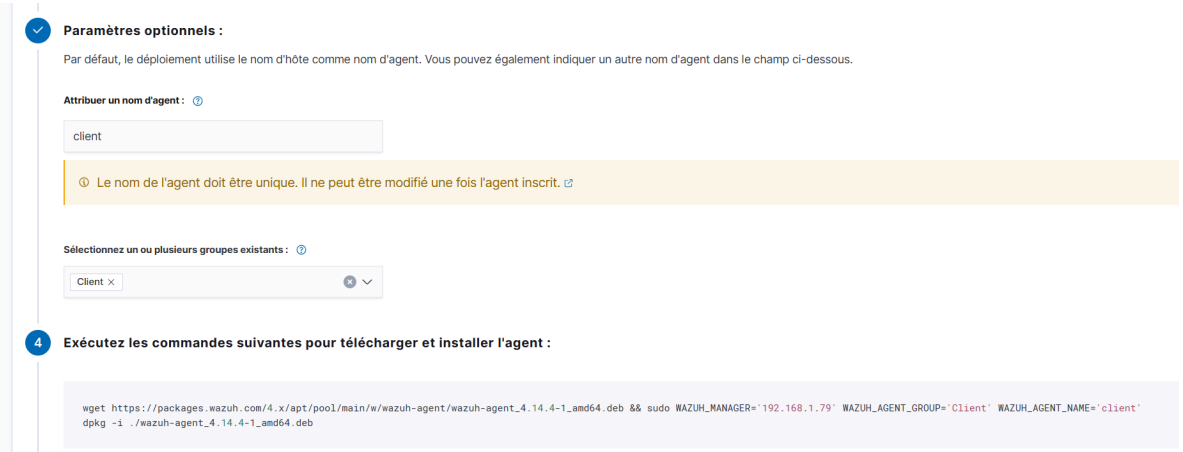
Je créer un groupe Client :



Je vais déployer un agent sur le client :







Exigences

- Vous aurez besoin de privilèges d'administrateur pour effectuer cette installation.
- Shell Bash est requis.

N'oubliez pas que vous devez exécuter cette commande dans un terminal Shell Bash.

5 **Démarrer l'agent :**

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

6 **Accédez aux points de terminaison pour vérifier la connexion de l'agent :**

Retour à la liste des agents

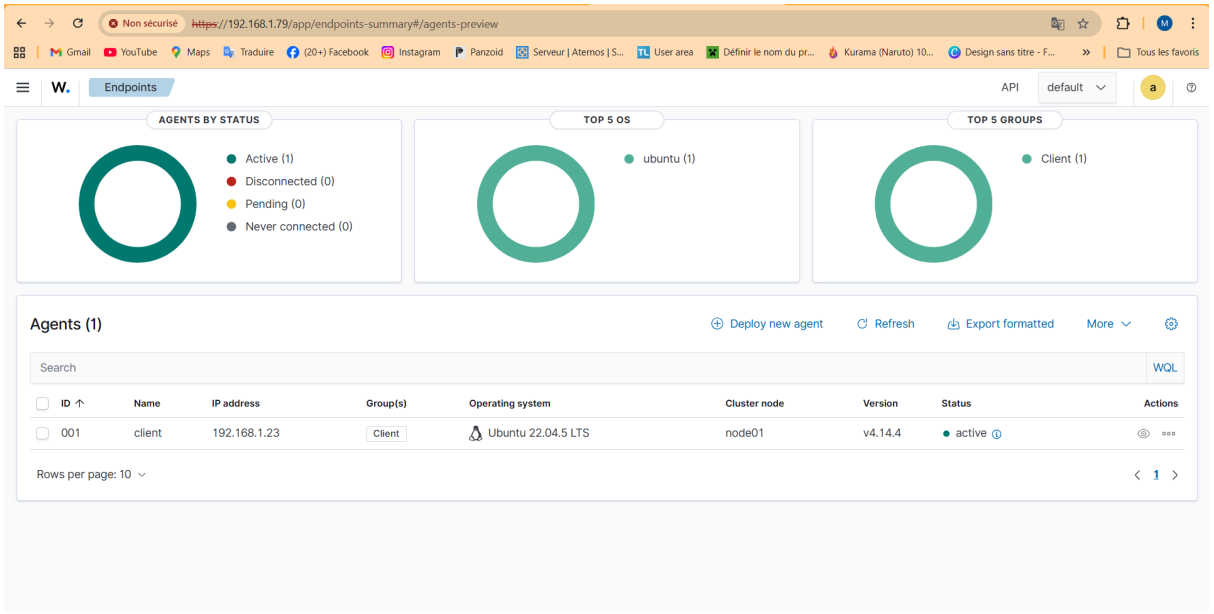
wget → télécharge le package
-4 → force IPv4 (corrige ton erreur)
WAZUH_MANAGER='192.168.1.79' → IP du serveur Wazuh
dpkg -i → installe l'agent

Il ne faut pas oublier le -4 sinon cela échouera car c'est en IPV6.

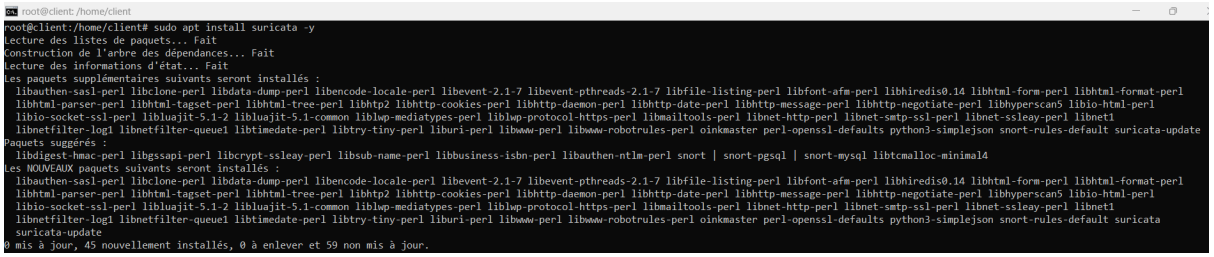
```
root@client: /home/client
root@client:/home/client# wget -4 https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.4-1_amd64.deb && sudo WAZUH_MANAGER='192.168.1.79' WAZUH_AGENT_GROUP='Client' WAZUH_AGENT_NAME='client' dpkg -i ./wazuh-agent_4.14.4-1_amd64.deb
--2026-03-26 20:43:58--  https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.4-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 3.162.38.84, 3.162.38.60, 3.162.38.122, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|3.162.38.84|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13220908 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.4-1_amd64.deb'

wazuh-agent_4.14.4-1_amd64.deb      100%[===================================================================================>]  12,61M  37,6KB/s    in 5m 28s

2026-03-26 20:49:28 (39,4 KB/s) - 'wazuh-agent_4.14.4-1_amd64.deb' saved [13220908/13220908]

Sélection du paquet wazuh-agent précédemment désélectionné.
(Lecture de la base de données... 75441 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../wazuh-agent_4.14.4-1_amd64.deb ...
Dépaquetage de wazuh-agent (4.14.4-1) ...
Paramétrage de wazuh-agent (4.14.4-1) ...
root@client:/home/client#
```

```
root@client:/home/client# sudo systemctl daemon-reexec
root@client:/home/client# sudo systemctl enable wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /lib/systemd/system/wazuh-agent.service.
root@client:/home/client# sudo systemctl start wazuh-agent
root@client:/home/client#
```

Agent Wazuh déployer sur le client :

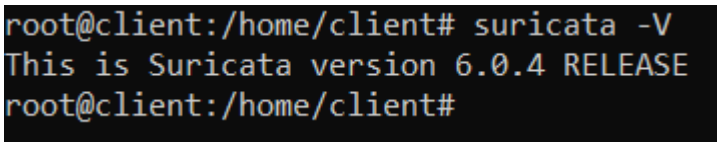


Installation de Suricata sur le client :



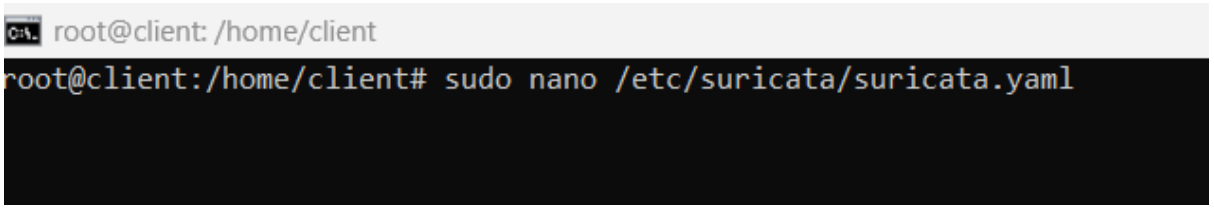
```
root@client: /home/client
root@client:/home/client# sudo apt install suricata -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  libauthen-sasl-perl libclone-perl libdata-dump-perl libencode-locale-perl libevent-2.1-7 libevent-pthreads-2.1-7 libfile-listing-perl libfont-afm-perl libhiredis0.14 libhtml-form-perl libhtml-format-perl
  libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl libhtp2 libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl libhttp-message-perl libhttp-negotiate-perl libhyperscan5 libio-html-perl
  libio-socket-ssl-perl libluajit-5.1-2 libluajit-5.1-common liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl libnet-ssleay-perl libnet1
  libnetfilter-log1 libnetfilter-queue1 libtimedate-perl libtry-tiny-perl liburi-perl libwww-perl libwww-robotrules-perl oinkmaster perl-openssl-defaults python3-simplejson snort-rules-default suricata-update
Paquets suggérés :
  libdigest-hmac-perl libgssapi-perl libcrypt-ssleay-perl libsub-name-perl libbusiness-isbn-perl libauthen-ntlm-perl snort | snort-pgsql | snort-mysql libtcmalloc-minimal4
Les NOUVEAUX paquets suivants seront installés :
  libauthen-sasl-perl libclone-perl libdata-dump-perl libencode-locale-perl libevent-2.1-7 libevent-pthreads-2.1-7 libfile-listing-perl libfont-afm-perl libhiredis0.14 libhtml-form-perl libhtml-format-perl
  libhtml-parser-perl libhtml-tagset-perl libhtml-tree-perl libhtp2 libhttp-cookies-perl libhttp-daemon-perl libhttp-date-perl libhttp-message-perl libhttp-negotiate-perl libhyperscan5 libio-html-perl
  libio-socket-ssl-perl libluajit-5.1-2 libluajit-5.1-common liblwp-mediatypes-perl liblwp-protocol-https-perl libmailtools-perl libnet-http-perl libnet-smtp-ssl-perl libnet-ssleay-perl libnet1
  libnetfilter-log1 libnetfilter-queue1 libtimedate-perl libtry-tiny-perl liburi-perl libwww-perl libwww-robotrules-perl oinkmaster perl-openssl-defaults python3-simplejson snort-rules-default suricata
  suricata-update
0 mis à jour, 45 nouvellement installés, 0 à enlever et 59 non mis à jour.
```

Vérifie :



```
root@client:/home/client# suricata -V
This is Suricata version 6.0.4 RELEASE
root@client:/home/client#
```

Modifier config :



```
root@client: /home/client
root@client:/home/client# sudo nano /etc/suricata/suricata.yaml
```

Fichier par défaut :

```
# Linux high speed capture support
af-packet:
  - interface: eth0
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
    # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
    # This is only supported for Linux kernel > 3.1
    # possible value are:
    #  * cluster_flow: all packets of a given flow are sent to the same socket
    #  * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
    #  * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
    #  socket. Requires at least Linux 3.14.
    #  * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
    #  more info.
    # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
    # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
    cluster-type: cluster_flow
    # In some fragmentation cases, the hash can not be computed. If "defrag" is set
    # to yes, the kernel will do the needed defragmentation before sending the packets.
    defrag: yes
    # To use the ring feature of AF_PACKET, set 'use-mmap' to yes
```

Remplace par TON interface :

```
# Linux high speed capture support
af-packet:
  - interface: enp0s3
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99
    # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
    # This is only supported for Linux kernel > 3.1
    # possible value are:
    #  * cluster_flow: all packets of a given flow are sent to the same socket
    #  * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
    #  * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
    #  socket. Requires at least Linux 3.14.
    #  * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
    #  more info.
    # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
    # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
    cluster-type: cluster_flow
    # In some fragmentation cases, the hash can not be computed. If "defrag" is set
    # to yes, the kernel will do the needed defragmentation before sending the packets.
    defrag: yes
```

Télécharger les règles :

```
root@client: /home/client
root@client:/home/client# sudo suricata-update
26/3/2026 -- 21:04:39 - <Info> -- Using data-directory /var/lib/suricata.
26/3/2026 -- 21:04:39 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
26/3/2026 -- 21:04:39 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
26/3/2026 -- 21:04:39 - <Info> -- Found Suricata version 6.0.4 at /usr/bin/suricata.
26/3/2026 -- 21:04:39 - <Info> -- Loading /etc/suricata/suricata.yaml
26/3/2026 -- 21:04:39 - <Info> -- Disabling rules for protocol http2
26/3/2026 -- 21:04:39 - <Info> -- Disabling rules for protocol modbus
26/3/2026 -- 21:04:39 - <Info> -- Disabling rules for protocol dnp3
26/3/2026 -- 21:04:39 - <Info> -- Disabling rules for protocol enip
26/3/2026 -- 21:04:39 - <Info> -- No sources configured, will use Emerging Threats Open
26/3/2026 -- 21:04:39 - <Info> -- Fetching https://rules.emergingthreats.net/open/suricata-6.0.4/emerging.rules.tar.gz.
   2% - 131072/5427298
```

Démarrer Suricata :

```
root@client: /home/client
root@client:/home/client# sudo systemctl enable suricata
Synchronizing state of suricata.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable suricata
root@client:/home/client# sudo systemctl start suricata
root@client:/home/client#
```

Vérifie :

```
root@client: /home/client
root@client:/home/client# sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
     Loaded: loaded (/lib/systemd/system/suricata.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-03-26 21:19:59 UTC; 35s ago
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata-ids.org/docs/
    Process: 4858 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid (code=exited, status=0/SUCCESS)
   Main PID: 4859 (Suricata-Main)
      Tasks: 8 (limit: 5085)
     Memory: 45.2M
        CPU: 12.974s
     CGroup: /system.slice/suricata.service
             └─4859 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid

mars 26 21:19:59 client systemd[1]: Starting Suricata IDS/IDP daemon...
mars 26 21:19:59 client suricata[4858]: 26/3/2026 -- 21:19:59 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
mars 26 21:19:59 client systemd[1]: Started Suricata IDS/IDP daemon.
root@client:/home/client#
```

Connecter Suricata à Wazuh :

Suricata écrit ses logs ici :

```
root@client: /home/client
root@client:/home/client# nano /var/log/suricata/eve.json
```

Ajouter dans Wazuh agent :

```
root@client: /home/client
root@client:/home/client# nano /var/ossec/etc/ossec.conf
```

Ajouté :

```
root@client: /home/client
  GNU nano 6.2                                                    /var/ossec/etc/ossec.conf *
  </localfile>

  <localfile>
    <log_format>full_command</log_format>
    <command>last -n 20</command>
    <frequency>360</frequency>
  </localfile>

  <!-- Active response -->
  <active-response>
    <disabled>no</disabled>
    <ca_store>etc/wpk_root.pem</ca_store>
    <ca_verification>yes</ca_verification>
  </active-response>

  <!-- Choose between "plain", "json", or "plain,json" for the format of internal logs -->
  <logging>
    <log_format>plain</log_format>
  </logging>



  <localfile>
    <log_format>journald</log_format>
    <location>journald</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/ossec/logs/active-responses.log</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/dpkg.log</location>
  </localfile>

<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>

</ossec_config>
```

Redémarre l’agent :

```
root@client: /home/client
root@client:/home/client# sudo systemctl restart wazuh-agent
root@client:/home/client#
```

Créer une règle de test personnalisée :

```
root@client: /home/client
root@client:/home/client# nano /var/lib/suricata/rules/local.rules
```

Ajoutez cette ligne (elle déclenche une alerte sur n'importe quel paquet ICMP/Ping) :

```
root@client: /home/client
  GNU nano 6.2                                                    /var/lib/suricata/rules/local.rules *
alert icmp any any -> any any (msg:"TEST WAZUH PING DETECTED"; sid:1000001; rev:1;)
```

Dis à Suricata de lire cette règle :

Ouvre ton fichier de configuration principal :

```
root@client: /home/client
root@client:/home/client# nano /etc/suricata/suricata.yaml
```

Vérifie que la section default-rule-path et rule-files inclut bien ton fichier :

Par défaut :

```
root@client: /home/client
  GNU nano 6.2                                                    /etc/suricata/suricata.yaml
    #    ports: [0-1,2-3,4-5,6-6,7-7]
    #
    # For legacy systems when hardware-bypass is disabled this can be specified in any
    # of the following ways:
    #
    #   a list of individual ports (e.g. ports: [0,1,2,3])
    #
    #   a range of ports (e.g. ports: [0-3])
    #
    #   "all" to indicate that all ports are to be merged together
    #   (e.g. ports: [all])
    #
    # This parameter has no effect if auto-config is disabled.
    #
    ports: [0-1,2-3]

    # When auto-config is enabled the hashmode specifies the algorithm for
    # determining to which stream a given packet is to be delivered.
    # This can be any valid Napatech NTPL hashmode command.
    #
    # The most common hashmode commands are:  hash2tuple, hash2tuplesorted,
    # hash5tuple, hash5tuplesorted and roundrobin.
    #
    # See Napatech NTPL documentation other hashmodes and details on their use.
    #
    # This parameter has no effect if auto-config is disabled.
    #
    hashmode: hash5tuplesorted

##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /etc/suricata/rules

rule-files:
  - suricata.rules

##
## Auxiliary configuration files.
##
```

Ajoute la nouvelle règle local.rules :

```
root@client: /home/client
  GNU nano 6.2                                                    /etc/suricata/suricata.yaml *
    ports: [0-1,2-3]

    # When auto-config is enabled the hashmode specifies the algorithm for
    # determining to which stream a given packet is to be delivered.
    # This can be any valid Napatech NTPL hashmode command.
    #
    # The most common hashmode commands are:  hash2tuple, hash2tuplesorted,
    # hash5tuple, hash5tuplesorted and roundrobin.
    #
    # See Napatech NTPL documentation other hashmodes and details on their use.
    #
    # This parameter has no effect if auto-config is disabled.
    #
    hashmode: hash5tuplesorted

##
## Configure Suricata to load Suricata-Update managed rules.
##

default-rule-path: /var/lib/suricata/rules

rule-files:
  - suricata.rules
  - local.rules
```

Redémarre Suricata :

```
root@client: /home/client
root@client:/home/client# systemctl restart suricata
root@client:/home/client#
```

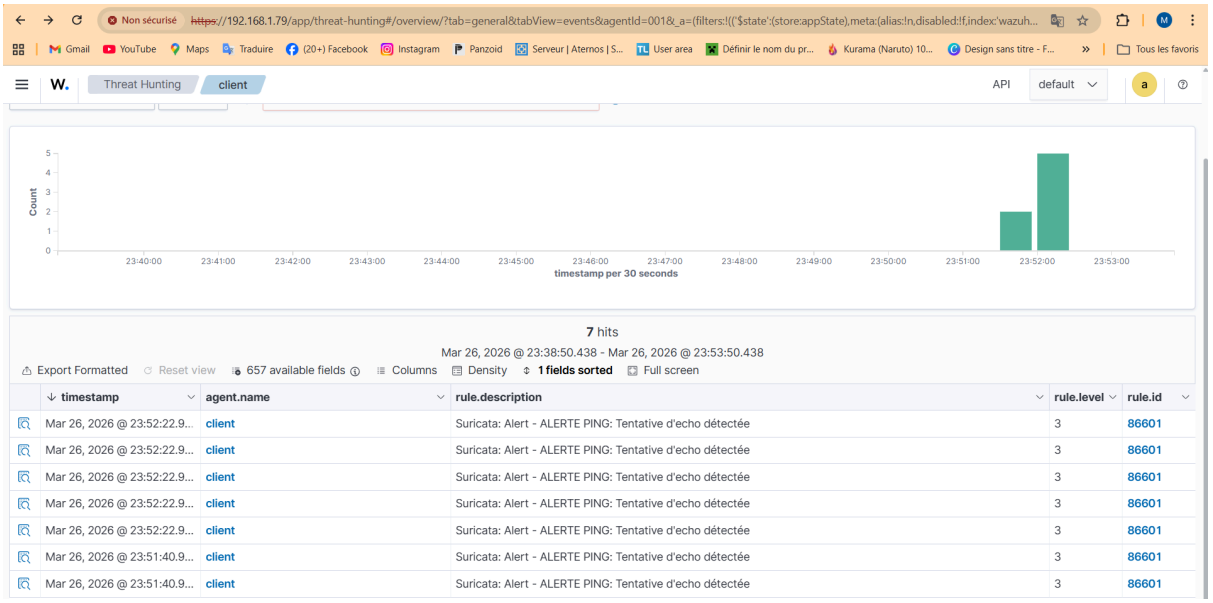
Valider la configuration :

```
root@client:/home/client# suricata -T -c /etc/suricata/suricata.yaml -v
26/3/2026 -- 22:28:26 - <Info> - Running suricata under test mode
26/3/2026 -- 22:28:26 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
26/3/2026 -- 22:28:26 - <Info> - CPUs/cores online: 2
26/3/2026 -- 22:28:26 - <Info> - fast output device (regular) initialized: fast.log
26/3/2026 -- 22:28:26 - <Info> - eve-log output device (regular) initialized: eve.json
26/3/2026 -- 22:28:26 - <Info> - stats output device (regular) initialized: stats.log
26/3/2026 -- 22:29:44 - <Info> - 2 rule files processed. 49226 rules successfully loaded, 0 rules failed
26/3/2026 -- 22:29:45 - <Info> - Threshold config parsed: 0 rule(s) found
26/3/2026 -- 22:29:48 - <Info> - 49231 signatures processed. 1272 are IP-only rules, 5215 are inspecting packet payload, 42527 inspect application layer, 108 are decoder event only
26/3/2026 -- 22:32:49 - <Notice> - Configuration provided was successfully loaded. Exiting.
26/3/2026 -- 22:32:50 - <Info> - cleaning up signature grouping structure... complete
root@client:/home/client#
```

Je lance un ping depuis Kali Linux :

```
┌──(root㉿carine)-[/home/carine]
└─# ping 192.168.1.23
PING 192.168.1.23 (192.168.1.23) 56(84) bytes of data.
64 bytes from 192.168.1.23: icmp_seq=1 ttl=64 time=2.60 ms
64 bytes from 192.168.1.23: icmp_seq=2 ttl=64 time=2.85 ms
64 bytes from 192.168.1.23: icmp_seq=3 ttl=64 time=3.67 ms
64 bytes from 192.168.1.23: icmp_seq=4 ttl=64 time=2.94 ms
^C
--- 192.168.1.23 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 2.604/3.015/3.667/0.395 ms
```

Depuis Wazuh nous allons voir l'alerte :




```
┌──(root㉿carine)-[/home/carine]
└─# nmap -sS 192.168.1.23
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-26 23:55 CET
Nmap scan report for client (192.168.1.23)
Host is up (0.0034s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 08:00:27:D9:1C:8E (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds
```

Non sécurisé https://192.168.1.79/app/threat-hunting#/overview/?tab=general&tabView=events&agentId=001&_a=(filters:!(),query:(language:kuery,query:suricata))&_g=(filters:!(),refreshI...

Gmail | YouTube | Maps | Traduire | (20+) Facebook | Instagram | Panzoid | Serveur | Aternos | S... | User area | Définir le nom du pr... | Kurama (Naruto) 10... | Design sans titre - F... | Tous les favoris

W. Threat Hunting / client — API — default — a

35 hits

Mar 26, 2026 @ 23:41:04.673 - Mar 26, 2026 @ 23:56:04.673

Export Formatted | Reset view | 657 available fields | Columns | Density | **1 fields sorted** | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 26, 2026 @ 23:56:03.1...	client	Suricata: Alert - ALERTE NMAP: Scan de ports détecté	3	86601
Mar 26, 2026 @ 23:55:47.1...	client	Suricata: Alert - ALERTE PING: Tentative d'echo détectée	3	86601
Mar 26, 2026 @ 23:55:47.1...	client	Suricata: Alert - ALERTE PING: Tentative d'echo détectée	3	86601
Mar 26, 2026 @ 23:55:45.2...	client	Suricata: Alert - ALERTE PING: Tentative d'echo détectée	3	86601
Mar 26, 2026 @ 23:55:45.2...	client	Suricata: Alert - ALERTE PING: Tentative d'echo détectée	3	86601